

MATHEMATICS

Time allotted: 3 hrs

Maximum Marks: 100

General Instructions:

- (i) All Questions are compulsory
- (ii) Please check that this question paper contains 26 questions
- (iii) Questions 1-6 in Section-A are very short-answer type questions carrying 1 mark each
- (iv) Questions 7-19 in Section-B are long-answer I type questions carrying 4 marks each
- (v) Questions 20-26 in Section-C are long-answer II type questions carrying 6 marks each
- (vi) Please write down the serial number of the question before attempting it

SECTION-A

Questions numbers 1-6 carry 1 mark each

1. If $x \in \mathbb{N}$ and $\begin{vmatrix} x+3 & -2 \\ -3x & 2x \end{vmatrix} = 8$, then find the value of x .
2. Use elementary column operation $C_2 \rightarrow C_2 + 2C_1$ in the following matrix equation:
$$\begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$
3. Write the number of all possible matrices of order 2×2 with each entry 1, 2 or 3.
4. Write the position vector of the point which divides the join of points with position vectors $3\vec{a} - 2\vec{b}$ and $2\vec{a} + 3\vec{b}$ in the ratio 2 : 1.
5. Write the number of vectors of unit length perpendicular to both the vectors $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{j} + \hat{k}$.
6. Find the vector equation of the plane with intercepts 3, -4 and 2 on x , y and z -axis respectively.

SECTION-B

Questions numbers 7-19 carry 4 marks each

7. Find the coordinates of the point where the line through the points $A(3, 4, 1)$ and $B(5, 1, 6)$ crosses the XZ plane. Also find the angle which this line makes with the XZ plane.
8. The two adjacent sides of a parallelogram are $2\hat{i} - 4\hat{j} - 5\hat{k}$ and $2\hat{i} + 2\hat{j} + 3\hat{k}$. Find the two unit vectors parallel to its diagonals. Using the diagonal vectors, find the area of the parallelogram.
9. In a game, a man wins Rs.5/- for getting a number greater than 4 and loses Rs.1/- otherwise, when a fair die is thrown. The man decided to throw a die thrice but to quit as and when he gets a number greater than 4. Find the expected value of the amount he wins/loses.

OR

A bag contains 4 balls. Two balls are drawn at random (without replacement) and are found to be white. What is the probability that all balls in the bag are white?

10. Differentiate $x^{\sin x} + (\sin x)^{\cos x}$ with respect to x .

OR

If $y = 2 \cos(\log x) + 3 \sin(\log x)$, prove that $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$.

11. If $x = a \sin 2t(1 + \cos 2t)$ and $y = b \cos 2t(1 - \cos 2t)$, find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$.
12. The equation of tangent at $(2, 3)$ on the curve $y^2 = ax^3 + b$ is $y = 4x - 5$. Find the values of a and b .
13. Find: $\int \frac{x^2}{x^4 + x^2 - 2} dx$